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BioContextAI is a community hub for agentic biomedical systems

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Large language models (LLMs) have recently been employed in biomedical research for diverse tasks, including cell-type annotation, literature mining and knowledge querying. Despite their potential applications, LLMs have limitations in biomedicine as they often generate hallucinations, cannot access specialized databases, and lack crucial domain knowledge and functional capabilities[1], which are essential to provide reliable scientific outputs. Recently, LLM-based agentic systems that do not solely rely on model-inherent knowledge have emerged. These systems autonomously use available tools to accomplish complex tasks[2].

Here, we introduce Biomedical Context for Artificial Intelligence (BioContextAI), an open-source initiative for creating agentic research assistants from standardized components (Fig. 1a). While web search integration has expanded LLM capabilities, this alone is insufficient to provide interconnected access to specialized resources, such as knowledgebases, medical literature, 'omics data and specialized tools, which are often behind unindexed application programming interfaces (APIs) or require additional software installation and file system access. Until now, this limitation has been tackled by retrieval-augmented generation[3], custom-developed function suites[4] or LLMs fine-tuned on biomedical data[5]. Despite achieving task-specific gains in accuracy, these solutions often lack reusability, interoperability and extensibility, conflicting with FAIR4RS[6] principles — an extension of FAIR data[7] principles to research software — and motivating more standardized approaches.

In late 2024, the Model Context Protocol (MCP) was introduced[8], aiming to address these challenges by providing a standardized protocol through which generative artificial intelligence systems can access predefined functions to interface with external information and applications. A key MCP concept is the decoupling of tool providers (servers) and LLM-powered tool consumers (clients or hosts), enabling flexible composability (Supplementary Fig. 1). This design allows agentic systems equipped with MCP support to extend their capabilities through a variety of MCP servers without additional development, thereby obviating the effort typically involved in integrating additional functionality (Supplementary Note 1).

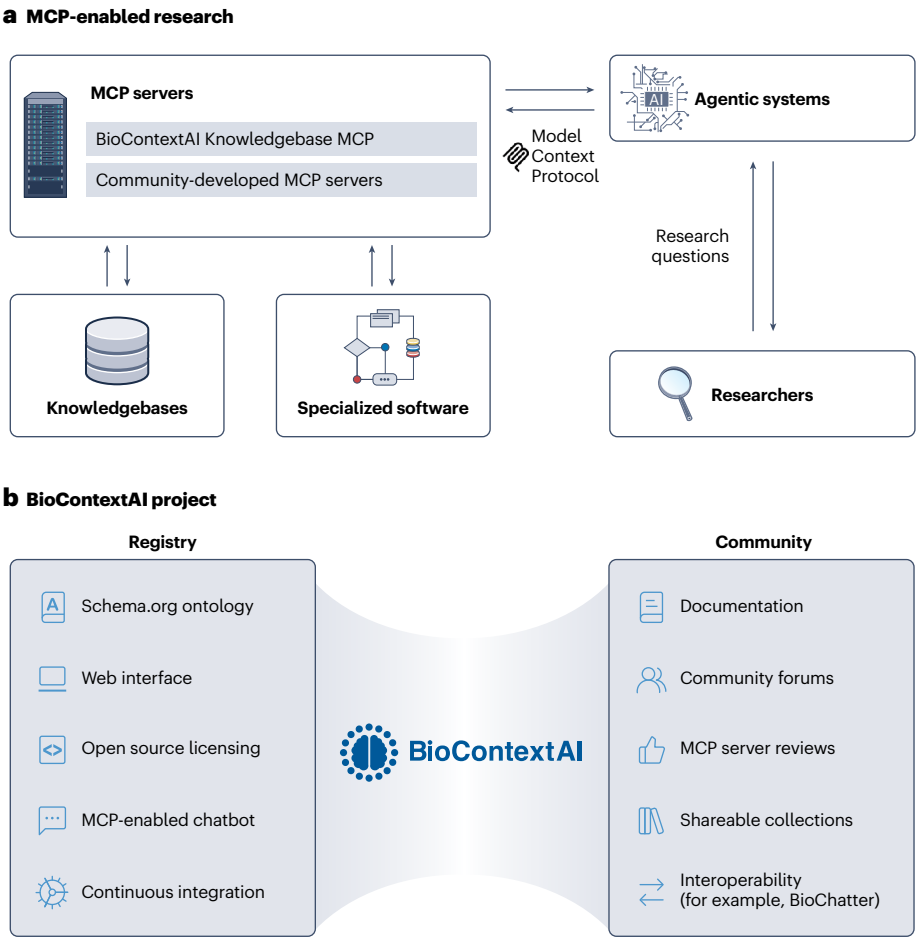

Fig. 1 | BioContextAI supports community-oriented agentic research in biomedicine. a, Facilitating access to specialized resources. In a Model Context Protocol (MCP)-enabled paradigm, investigators can pose their research questions in natural language. LLM-based agentic systems then query MCP servers to access information from knowledgebases or to leverage specialized software. **b**, Toward a modular MCP server ecosystem. BioContextAI indexes community-built MCP servers in its Registry with Schema.org ontology, an accessible web interface and related tooling. This fosters a community of MCP server developers and the researchers who use them, offering extensive documentation, forums, a review system and shareable collections to help users choose adequate MCP servers for their scientific endeavors.

MCP partially mitigates FAIR4RS challenges by standardizing the protocol for tool discovery and interaction, but substantial gaps remain. There is still no community-based registry that offers comprehensive, searchable metadata, so the broad use of MCP servers is limited. Missing or overly restrictive licenses, limited metadata and unclear long-term support of existing servers further hamper their accessibility. Additionally, varying

documentation quality and unclear origins of data and code impact reusability.

To address these issues, BioContextAI adopts a standard-oriented, open-source community model inspired by initiatives like the scverse[9] and BioCypher[10] (Supplementary Methods). At its core sits the Registry (Fig. 1b), which leverages Schema.org ontology terms[11] to catalog community-developed MCP servers in a public repository with rich metadata while enforcing documentation and open-source licensing standards. These MCP servers alleviate the burden on developers of agentic systems and unlock synthesis and recombination potential by eliminating the need to reimplement the layers through which LLMs access domain-specific resources.

MCP server developers can use an interactive online editor to seamlessly create metadata files to add their MCP server to the Registry, contributing it through the Registry's GitHub repository (Supplementary Fig. 2). Researchers can interact with the Registry through a downloadable index, a REST API or the interactive web interface available at https://biocontext.ai/. In addition, the community can identify the most helpful MCP servers for different use cases through a review system and shareable collections. To facilitate the exploration of biomedical MCP servers, the web application provides access to BioContextAI Chat, a chatbot capable of connecting to any remote MCP server in our Registry, offering an accessible way of trying out remote MCP servers. Furthermore, online documentation provides useful resources for MCP server development and usage tips for BioContextAI. As a practical starting point, a feature-rich MCP server cookiecutter template further streamlines the development of new MCP servers in accordance with BioContextAI's standards.

As a foundation for full-featured MCP server development and as an interactive example, BioContextAI offers the Knowledgebase MCP. This MCP server implementation exposes functionality from over 15 biomedical resources, including UniProt[12] and Open Targets[13] (Supplementary Note 2). It is available as an open-source software package for local deployment through the PyPI Python software registry, enabling flexible combination with existing MCP servers to enhance agentic systems with extended biomedical capabilities. A remote-hosted version can be integrated setup-free across an array of generative artificial intelligence applications.

Despite these advances, challenges persist. Reliance on commercial LLMs and use of remote-hosted MCP servers present security, authentication, financial, data privacy and maintenance concerns[14], which could put them out of reach for certain research applications. Future developments in hybrid local–cloud architectures, community-driven governance models and privacy-preserving computation will be vital to overcoming these limitations.

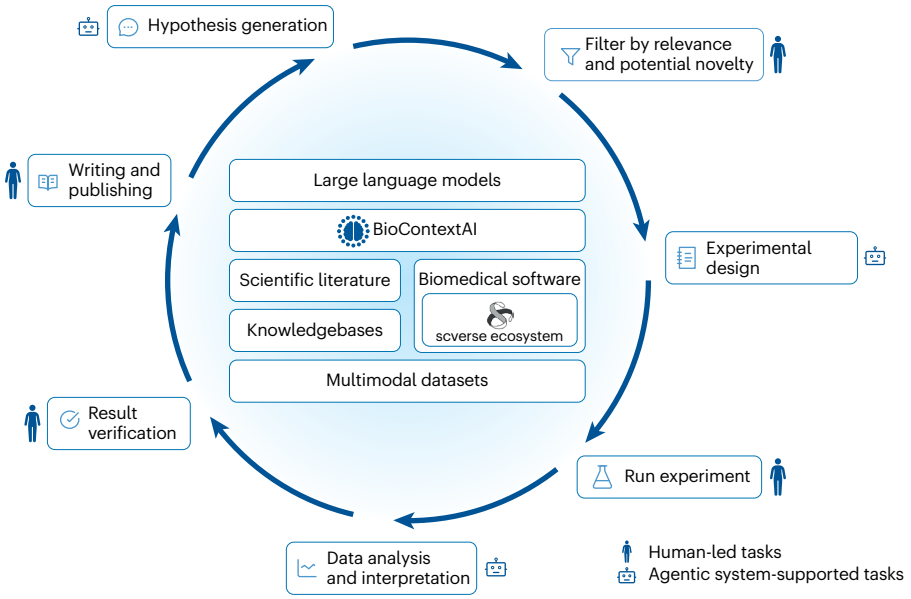

Fig. 2 | Toward reliable agentic system supported biomedical research. Agentic systems, relying on BioContextAI to find and develop MCP servers, integrate scientific literature, knowledgebases, biomedical software (for example, scverse ecosystem packages) and multimodal datasets. In an iterative agent–human research loop, these systems could be used to support key stages, such as hypothesis generation, experimental design and data analysis. Human researchers would assume a more supervisory role, ensuring relevance, validity of results and accurate communication of findings to the wider scientific community.

In this light, BioContextAI is a core component of an emerging open-source ecosystem of agentic systems for biomedical research. Given the rapid progress of open-weights reasoning LLMs and optimizations such as quantization and mixture-of-experts architectures[15], combined with improvements in hardware throughput, the deployment of privacy-preserving, performant, transparent and reproducible agentic systems in academic research is becoming increasingly feasible.

Looking ahead, agent-supported research may involve systems that operate beyond knowledge retrieval and standardized data analysis. Future systems may touch on all phases of scientific discovery (Fig. 2), autonomously researching literature to identify gaps in current knowledge, aiding in experimental design, composing analysis code, supporting analysis by enhancing outputs with data obtained from knowledgebases and biomedical software, and incorporating findings into an ever-expanding body of biomedical literature. Importantly, human researchers must stay in the loop to verify the accuracy of outputs and ensure that proposed experiments are both innovative and scientifically sound. Equitable access to these emerging technologies will be key to leveraging existing expertise and ensuring that developments reflect the richness of the scientific community and its members.

We are committed to the continued development of BioContextAI as a community hub for LLM-supported biomedical research. As additional MCP servers are developed, a growing collection of biomedical tools with relevant technical capabilities will be made broadly accessible through the Registry. Researchers and developers will be supported in creating, evaluating, and deploying MCP-augmented agentic systems through common standards, best-practice guides, templates, and integration tests. To enhance visibility and engagement and strive for an interoperable ecosystem, BioContextAI will partner with existing communities, including BioCypher and the scverse, and use open communication channels to foster exchange between users and developers. Finally, researchers and developers can begin to engage with BioContextAI

by interacting with the community, contributing MCP servers, developing agentic systems and integrating available tools into research workflows to help advance the understanding of health and disease.

Code availability

The BioContextAI Registry is available at https://github.com/biocontext-ai/registry. A searchable index and tutorial resources are available at https://biocontext.ai. The BioContextAI Knowledgebase MCP code is available at https://github.com/biocontext-ai/knowledgebase-mcp. It can be installed from PyPI as "biocontext-kb". A remote-hosted version is available with streamable HTTP transport at https://mcp.biocontext.ai/mcp/ (subject to usage restrictions). The API documentation is available at https://docs.kb.biocontext.ai. BioContextAI Chat is available as a freely usable chat interface at https://biocontext.ai/chat. Source code for the BioContextAI website – including the Registry interface and chat – is available at https://github.com/biocontext-ai/website. The evaluation scripts can be accessed at https://github.com/biocontext-ai/simple-mcp-evaluation. The MCP server cookiecutter template is available at https://github.com/biocontext-ai/mcp-server-cookiecutter.

Malte Kuehl[1,2,19] ✉, **Darius P. Schaub**[3,4,19], **Francesco Carli**[5,6], **Lukas Heumos**[7], **Malte Hellmig**[4,8,9], **Camila Fernández-Zapata**[4,8], **Nico Kaiser**[3,4,8], **Jonathan Schaul**[10], **Anton Kulaga**[11], **Nikolay Usanov**[12], **Scverse Community***, **Christian F. Krebs**[4,8,9], **Ulf Panzer**[4,8,9], **Stefan Bonn**[3,8,9,13,20], **Sebastian Lobentanzer**[5,7,14,20], **Julio Saez-Rodriguez**[5,6,10,20] & **Victor G. Puelles**[1,2,4,8,20] ✉

[1]Department of Clinical Medicine, Aarhus University, Aarhus, Denmark. [2]Department of Pathology, Aarhus University Hospital, Aarhus, Denmark. [3]Institute of Medical Systems Bioinformatics, Center for Biomedical AI (bAIome), Center for Molecular Neurobiology (ZMNH), University Medical Center Hamburg-Eppendorf, Hamburg, Germany. [4]III. Department of Medicine, University Medical Center Hamburg-Eppendorf, Hamburg, Germany. [5]Open Targets, European Bioinformatics Institute (EMBL-EBI), Cambridge, UK. [6]European Molecular Biology Laboratory, European Bioinformatics Institute (EMBL-EBI), Hinxton, UK. [7]Institute of Computational Biology, Computational Health Center, Helmholtz Center, Munich, Germany. [8]Hamburg Center for Kidney Health (HCKH), University Medical Center Hamburg-Eppendorf, Hamburg, Germany. [9]Hamburg Center for Translational Immunology (HCTI), University Medical Center Hamburg-Eppendorf, Hamburg, Germany. [10]Institute for Computational Biomedicine, Heidelberg University, Faculty of Medicine, Heidelberg University Hospital, Heidelberg, Germany. [11]Institute for Biostatistics and Informatics in Medicine and Ageing Research, Rostock University Medical Center, Rostock, Germany. [12]HEAlthy Life Extension Society (HEALES), Brussels, Belgium. [13]German Center for Child and Adolescent Health (DZKJ), partner site Hamburg, University Medical Center Hamburg-Eppendorf, Hamburg, Germany. [14]Computational Biology Unit, German Center for Diabetes Research, Munich, Germany. [19]These authors contributed equally: Malte Kuehl, Darius P. Schaub. [20]These authors jointly supervised this work: Stefan Bonn, Sebastian Lobentanzer, Julio Saez-Rodriguez, Victor G. Puelles. *A list of authors and their affiliations appears at the end of the paper.

✉e-mail: malte.kuehl@clin.au.dk; vgpuelles@clin.au.dk

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Author contributions

M.K., D.P.S. and V.G.P. conceived the study. M.K. wrote the first draft. M.K. and D.P.S. developed the BioContextAI Knowledgebase MCP package, the Registry, BioContextAI Chat and the MCP server evaluation, with F.C., S.L., J.S., C.F.-Z., N.K., A.K. and N.U. providing inputs. M.H., M.K. and D.P.S. developed the MCP cookiecutter template. F.C. and S.L. implemented the BioChatter integration. M.K., L.H., D.P.S., F.C., the Scverse Community, S.L., J.S.-R. and V.G.P. developed the community structure. M.K. and V.G.P. created the figures. C.F.K., U.P., S.B., S.L., J.S.-R. and V.G.P. acquired funding. S.B., S.L., J.S.-R. and V.G.P. supervised the study. All authors contributed to the editing of and approved the final draft.

Competing interests

M.K. is an employee of and holds an ownership interest in KH Biotechnology, which provides consulting services to Lamin Labs. L.H. is an employee of Lamin Labs. J.S.-R. reports in the last 3 years funding from GSK and Pfizer and fees or honoraria from Travere Therapeutics, Stadapharm, Astex, Owkin, Pfizer, Grunenthal, Tempus and Moderna.

Additional information

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Scverse Community

Mikaela Koutrouli[15], **Can Ergen**[16,17] & **Giovanni Palla**[18]

[15]Genentech, San Francisco, CA, USA. [16]Department of Electrical Engineering and Computer Sciences, University of California, Berkeley, Berkeley, CA, USA. [17]University Hospital Hamburg-Eppendorf, I. Medical Department, University Medical Center Hamburg-Eppendorf, Hamburg, Germany. [18]Chan Zuckerberg Initiative, Redwood City, CA, USA.